

# MATH 345 Skeleton Notes

## Unit 1: Vectors, matrices, and linear maps

### Section 1.6: Exercises

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*Exercise tags.* The tags below identify the primary outcomes and competency types for each exercise group. Individual problems may also involve earlier outcomes.

#### Matrices

*Primary outcomes.* U1-LO3. *Competencies.* P.

- Nicholson 2.1.3(f)
- Nicholson 2.1.8(b)
- Nicholson 2.1.15(b)

#### Matrix-vector multiplication

*Primary outcomes.* U1-LO4. *Competencies.* P+R.

- Nicholson 2.2.2(b)
- Nicholson 2.2.3(b)
- Nicholson 2.2.11(d)

#### Linear and affine maps

*Primary outcomes.* U1-LO5, U1-LO7. *Competencies.* P+C+R.

- Nicholson 2.6.2(b)
- Nicholson 2.6.13(b)

## Matrix multiplication

*Primary outcomes.* U1-LO6. *Competencies.* P+R.

- Nicholson 2.3.1(b)
- Nicholson 2.3.3(b)
- Nicholson 2.3.5(b)
- Nicholson 2.3.16(b)

## Additional applied and computational problems

These exercises connect the main Unit 1 ideas: vectors as data, cosine similarity, weighted averages, matrix-vector products, geometric matrix actions, composition, affine maps, and code interpretation. Solutions are collected in Appendix Unit 1 applied, geometric, and computational interpretation.

### Exercise: Word-count vectors and cosine similarity

Use the dictionary `linear`, `matrix`, `data`. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

1. Compute the cosine similarity between  $\mathbf{q}$  and each document vector.

2. Rank the documents by cosine similarity.
3. Explain why  $\mathbf{D}_1$  has cosine similarity 1 with  $\mathbf{q}$ .
4. Which document is orthogonal to  $\mathbf{q}$ ?
5. What does that orthogonality mean in terms of the dictionary words?
6. Compute the Euclidean distance from  $\mathbf{q}$  to each  $\mathbf{D}_i$ . Does the distance ranking match the cosine ranking?

*Tags.* [U1-L01, U1-L02 | C+R | Core]

### **Exercise: Weighted average of value vectors**

Let

$$\mathbf{v}_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 10 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 10 \\ 10 \end{bmatrix},$$

and let

$$\alpha = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2} \end{bmatrix}.$$

1. Compute  $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$ .
2. Explain why this is a weighted average.
3. Which value vector receives the largest weight?

*Tags.* [U1-L08 | P+C | Core]

### **Exercise: Attention bridge: hand scores versus $K\mathbf{q}$**

Use the token attention activity with tokens “small”, “red”, “bird”. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three dot products

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}.$$

2. Put the key vectors as rows of the matrix

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Compute  $K\mathbf{q}$ .

3. Explain why the entries of  $K\mathbf{q}$  are the same scores from part (a).
4. Normalize the scores by dividing by their sum. What is the weight vector  $\alpha$ ?

*Tags.* [U1-L01, U1-L04, U1-L08 | C+R | Core]

### **Exercise: Rows measure; columns contribute**

Let

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

1. Compute  $A\mathbf{x}$  using row dot products.
2. Compute  $A\mathbf{x}$  as a linear combination of the columns of  $A$ .

3. In this example, what does the first row of  $A$  measure?
4. In this example, what do the coordinates  $2, -3, 4$  tell you in the column-combination view?

Tags. [U1-L04 | P+R | Core]

### Exercise: The unit-square visualization

Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}.$$

1. Compute  $A\mathbf{e}_1$  and  $A\mathbf{e}_2$ .
2. Compute  $A \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .
3. Apply  $A$  to the four corners of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Sketch the image of the unit square.
5. Which two transformed edges leaving the origin are the columns of  $A$ ?

Tags. [U1-L05, U1-L08 | C+R | Core]

### Exercise: Transformation matching

Match each matrix with the correct geometric description.

$$\begin{aligned} A_1 &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, & A_2 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, & A_3 &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \\ A_4 &= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, & A_5 &= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, & A_6 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

1. Reflection across the  $y$ -axis.
2. Projection onto the  $x$ -axis.
3. Horizontal stretch.
4. Horizontal shear.
5.  $90^\circ$  counterclockwise rotation.
6. Reflection across the line  $y = x$ .

For each match, briefly justify your answer by describing what happens to  $\mathbf{e}_1$  and  $\mathbf{e}_2$ .

Tags. [U1-L05 | C+R | Core]

### Exercise: Composition order: $RS$ versus $SR$

Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute  $RS$ .
2. Compute  $SR$ .
3. Are  $RS$  and  $SR$  equal?
4. Which product represents “apply  $S$  first, then apply  $R$ ”?
5. Explain geometrically why the two orders produce different results.

Tags. [U1-L06 | P+R | Core]

### Exercise: Affine but not linear

Consider the transformation

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 1 \\ y \end{bmatrix}.$$

1. Compute  $T(0, 0)$ .
2. Explain why  $T$  cannot be written in the form  $T(\mathbf{x}) = A\mathbf{x}$  for any matrix  $A$ .
3. Write  $T$  in the affine form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}.$$

4. Is  $T$  linear, affine but not linear, or neither?

*Tags.* [U1-L07 | C+R | Core]

### **Exercise: Code interpretation: sorting similarity scores**

Consider the code:

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])
scores = np.array([1.0, 0.5, 0.0])
ranking = doc_names[np.argsort(scores)::-1]
ranking
```

1. What is the value of `ranking`?
2. What does `np.argsort(scores)` return?
3. What does `::-1` do?
4. What is being ranked?
5. Is this ranking based on direction, distance, or alphabetical order?

*Tags.* [U1-L02, U1-L08 | R+T | Core]

### **Exercise: Transformer shape check**

Suppose a sequence of token vectors is stored in a matrix

$$X \in \mathbb{R}^{6 \times 5}.$$

Let

$$W_Q, W_K, W_V \in \mathbb{R}^{5 \times 2},$$

and define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A later weighting step produces an attention-weight matrix  $A_{\text{att}}$ , and the attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of  $Q$ ,  $K$ , and  $V$ ?
2. What is the shape of  $K^T$ ?
3. What is the shape of  $S = QK^T$ ?
4. What is the shape of  $A_{\text{att}}$ ?
5. What is the shape of  $H$ ?
6. What does the  $(i, j)$ -entry of  $S$  compute?
7. Which product forms the weighted averages of value vectors?
8. In this convention, are token vectors stored as rows or columns of  $X$ ?

*Tags.* [U1-L03, U1-L06, U1-L08 | R+T | Core]